This assignment will not be graded and is only for practice.

1) Consider a map $T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$ defined as $T(v) = Av$, where $v = \begin{bmatrix} x \\ y \end{bmatrix}$, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $det(A) = 0$. Which of the following options are correct? ***1 point***

- [ ] T is a linear transformation.
- [ ] T is both one-one and onto.
- [ ] T is neither one-one nor onto.
- [ ] T is one-one but not onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is a linear transformation.
T is both one-one and onto.

Recall the definition of a linear transformation

- A function T from one vector space V to another vector space W ($T : V \longrightarrow W$) is said to be a linear transformation if for any two vectors v_1 and v_2 in the vector space V and for any $c \in \mathbb{R}$ (scalar) the following conditions hold:
(i) $T(v_1 + v_2) = T(v_1) + T(v_2)$
(ii) $T(cv_1) = cT(v_1)$

Step 1:

2) What is $T(v_1 + v_2)$? ***1 point***

- ( ) $A(v_1 + v_2)$.

- ○ Av_1v_2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$A(v_1 + v_2)$.

Recall: $A(v_1 + v_2) = Av_1 + Av_2$

Step 2:

3) Is $T(v_1 + v_2) = T(v_1) + T(v_2)$? ***1 point***

- ○ Yes.
- ○ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

Check: $T(cv) = cT(v)$.

Hence T is linear transformation. So Option 1 is true.

Another method: Write the definition of T explicitly as,

$$T(x, y) = (ax + by, cx + dy)$$

Step 1: What is $T((x_1, y_1) + (x_2 + y_2))$?

$$\begin{aligned} T(x_1 + x_2, y_1 + y_2) &= (a(x_1 + x_2) + b(y_1 + y_2), c(x_1 + x_2) + d(y_1 + y_2)) \\ &= (ax_1 + ax_2 + by_1 + by_2, cx_1 + cx_2 + dy_1 + dy_2) \\ &= (ax_1 + by_1, cx_1 + dy_1) + (ax_2 + by_2, cx_2 + dy_2) \\ &= T(x_1, y_1) + T(x_2, y_2) \end{aligned}$$

Step 2:

4) Is $T(c(x, y)) = cT(x, y)$? ***1 point***

- ○ Yes.
- ○ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

Hence T is linear transformation. So Option 1 is true.

Recall the definition of a one to one and an onto linear transformation $T : V \longrightarrow W$

- If $T(v_1) = T(v_2)$ implies $v_1 = v_2$, then T is one to one.
- If for each $w \in W$ there exists $v \in V$ such that $T(v) = w$, then T is onto.
- $T(v) = 0$ implies $v = 0$, if and only if T is one to one.
  - Suppose T is one to one. Now let us assume that $T(v) = 0$. We also have $T(0) = 0$. Hence, $T(v) = T(0)$. As T is one to one, by the definition of a one to one transformation, we have $v = 0$.
  - Conversely assume that $T(v) = 0$ implies $v = 0$. We will show T is one to one. Let $T(v_1) = T(v_2)$. So we have, $T(v_1 - v_2) = T(v_1) - T(v_2) = 0$. Hence $v_1 - v_2 = 0$, which implies $v_1 = v_2$. So T is one to one.

Now in the given problem $T(v) = 0$, implies $Av = 0$. This is the matrix representation of the system of linear equations:

$$ax + by = 0$$

$$cx + dy = 0$$

Step 3:

5) If the system of linear equations $Av = 0$ has a unique solution, then what is the solution? ***1 point***

- ○ $v = 0$
- ○ v can be any vector in $\mathbb{R}^2$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$v = 0$

Step 4:

6) What is the condition on A, for which the system of linear equations $Av = 0$ has a unique solution? [Recall from Weeks 1 and 2] **1 point**

- $det(A) = 0.$
- $det(A) = 0.$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$det(A) = 0.$

Concluding, we can say that, if $det(A)$ is non-zero, then $T(v) = 0$ implies $v = 0$. Hence T is one to one.

Let $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \in \mathbb{R}^2$. We are going to find whether there exists a vector $v = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$, such that $T(v) = Av = w$.

$Av = w$ is the matrix representation of the system of linear equation:

$$ax + by = w_1$$
$$cx + dy = w_2$$

Step 5:

7) If $det(A) = 0$, then what can we say about the solution of the above system of linear equations? **1 point**

- No solution.
- Unique solution.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Unique solution.

Concluding from this, we can say that, if $det(A) = 0$, then T is onto.

Hence, Options 1 and 2 are the correct options.

Your score is: 0/7